

Salesforce Interview Readiness Bootcamp

Day 3 – Validation Rules, Flows & Triggers

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Objective

The objective of Day 3 was to understand Salesforce automation using Validation Rules, Record-Triggered Flows, and Apex Triggers. The main focus was learning how to choose the right automation tool based on different business requirements and implementing automation in the Placement Management System.

Topics Covered

- Validation Rules
- Flow Builder
- Record-Triggered Flow
- Before Save Flow
- After Save Flow
- Apex Trigger Basics
- Flow vs Trigger
- Order of Execution (Basics)

Concepts Learned

Validation Rule

Validation Rules are used to stop users from saving invalid data. They only validate the data and display an error message if the condition is not met.

Example: Student CGPA should be greater than or equal to the minimum CGPA required for a job.

Flow

Flow is Salesforce's declarative automation tool. It allows us to automate business processes without writing Apex code.

Some common uses are:

- Updating fields
- Creating records
- Sending emails
- Updating related records

Apex Trigger

Triggers are written using Apex and are used when business logic becomes too complex for Flow.

Triggers are useful for:

- Complex calculations
- Bulk record processing

- Advanced business logic
-

Business Scenario

The Placement Cell wanted to automate the application process.

Requirements:

1. Automatically fill the Application Date.
 2. Send an email when a student applies.
 3. Prevent duplicate applications.
 4. Reject students with low CGPA.
 5. Create an Offer Letter when the student is selected.
-

Objects Used

The following custom objects were used:

- Student
- Job
- Application
- Offer Letter

Fields Added

Student

- Student Name
- Email
- CGPA
- Department

Job

- Job Name
- Company
- Minimum CGPA
- Closing Date

Application

- Student (Lookup)
- Job (Lookup)
- Application Date
- Status

Offer Letter

- Application (Lookup)
 - Offer Date
 - Status
-

Task 1 – Before Save Record-Triggered Flow

Purpose

Automatically populate the Application Date whenever a new Application record is created.

Configuration

- Flow Type: Record-Triggered Flow
- Object: Application
- Trigger: Record Created
- Optimization: Fast Field Updates (Before Save)

Logic

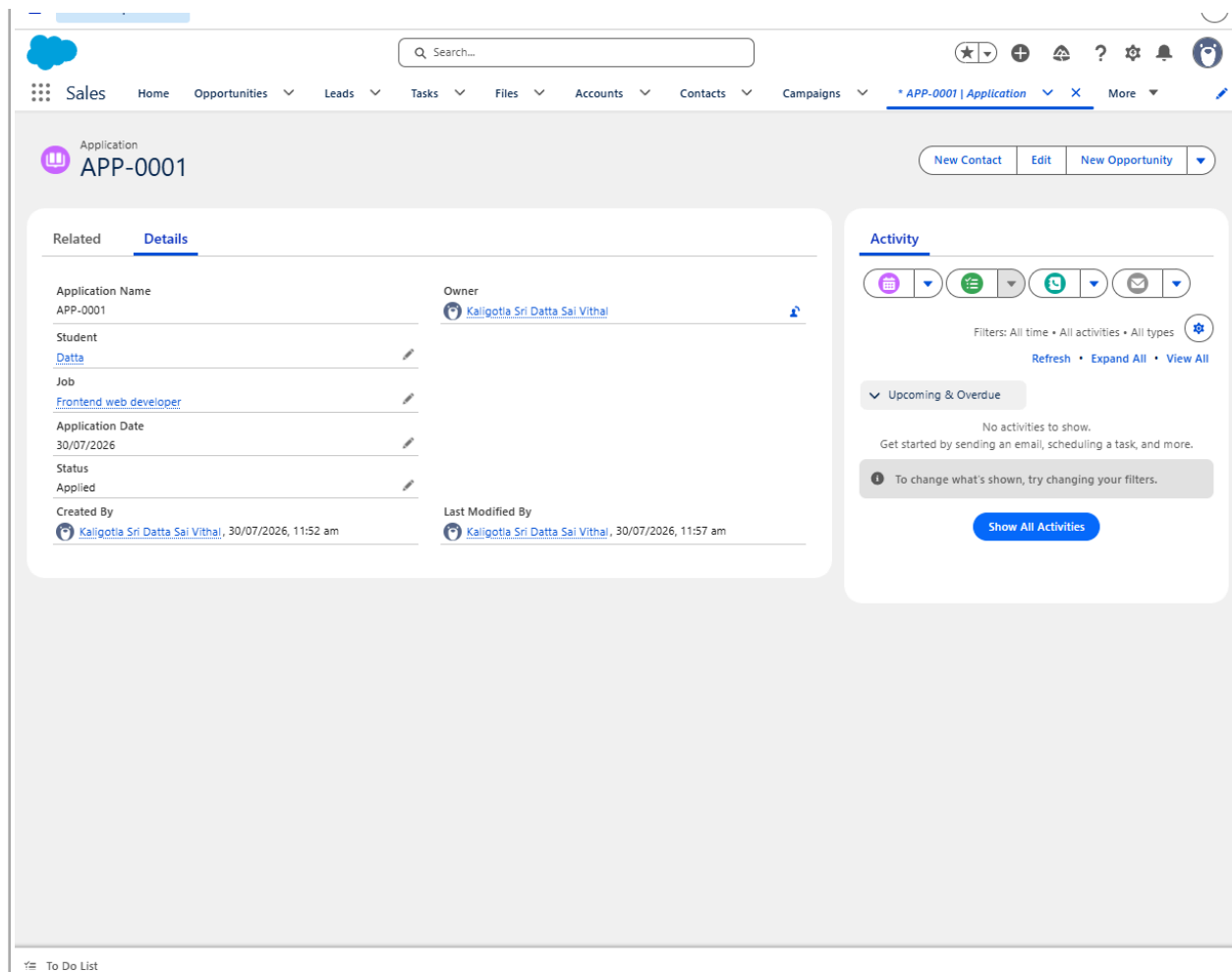
Application Date = \$Flow.CurrentDate

Result

Whenever a new Application record was created, Salesforce automatically filled today's date in the Application Date field.

The screenshot displays the Salesforce Flow Builder interface. The top navigation bar shows 'Developer Edition' and the flow name 'Application Before Save Flow - V1'. The flow canvas on the left shows a sequence of steps: 'Record-Triggered Flow Start', 'Set Application Date' (Assignment), and 'End'. The right sidebar provides configuration details for the 'Set Application Date' step. It includes a 'Label' field with the text 'Set Application Date', an 'API Name' field with the text 'Set_Application_Date', and a 'Description' field. Below these, the 'Set Variable Values' section shows a variable assignment: '...application__c > Application Date' is set equal to '...ng Flow Interview > CurrentDate'.

Flow Canvas



task1-flow-canvas.png

Task 2 – After Save Record-Triggered Flow

Purpose

Send an email notification whenever a new Application record is created.

Configuration

- Flow Type: Record-Triggered Flow
- Object: Application
- Trigger: Record Created
- Optimization: Actions and Related Records (After Save)

Email Details

Recipient: Placement Officer (used my email for testing)

Subject: New Application Submitted

Email Body

Hello Placement Officer,

A new application has been submitted.

Student: {!\$Record.Student__r.Name}

Job: {!\$Record.Job__r.Name}

Status: {!\$Record.Status__c}

Please log in to Salesforce to review the application.

Regards,
Salesforce Automation

Result

Whenever a new Application record was created, an email notification was automatically sent.

The screenshot displays the Salesforce Flow Builder interface. The main canvas shows a flow diagram with the following steps:

- Record-Triggered Flow** (Start): Object: **Application**, Trigger: **A record is created**, Optimize for: **Actions and Related Records**.
- Run Immediately** (Intermediate step).
- Send Email** (Action): Configured to send an email.
- End** (Final step).

The right-hand panel shows the configuration for the **Send Email** action:

- Configure Email Content**:
 - Compose Email Content** (Selected): Set email details and compose message.
 - Use Email Template**: Select an existing email template.
- Subject**: New Application Submitted
- Body**:
 - Text: Hello Placement Officer,
 - Text: A new application has been submitted.
 - Text: Student: {!\$Record.Student__r.Name}
 - Text: Job: {!\$Record.Job__r.Name}
 - Text: Status: {!\$Record.Status__c}
 - Text: Please login to Salesforce to review the application.

Email Action

Validation Rules Created

Created Validation Rules for:

1. Student CGPA must meet the minimum CGPA required for the Job.

2. Application Date should not be after the Job Closing Date.
3. Required fields should not be left blank.



Paste screenshot here: *Validation Rule Setup*

(suggested filename: validation-rule-setup.png)



Paste screenshot here: *Validation Error*

(suggested filename: validation-error.png)

Flow vs Trigger

Requirement	Tool Used	Reason
Auto-fill Application Date	Flow	Simple field update
Send Email	Flow	Built-in email action
Reject Low CGPA	Validation Rule	Prevent invalid data
Prevent Duplicate Application	Validation Rule	Stop duplicate records
Complex Business Logic	Trigger	When Flow is not enough

Testing

I tested the automation by creating new Application records.

Test 1

Created a new Application without entering the Application Date

Result: Application Date was automatically populated..

Developer Edition

Flow Builder Application Before Save Flow - V1

Last saved on 30/7/2026, 11:45 am **Active** Run Debug View Tests Save As New Version Save Deactivate

Auto-Layout

```

graph TD
    Start([Record-Triggered Flow Start]) --> Assignment[Set Application Date Assignment]
    Assignment --> End([End])
  
```

Assignment

*Label: Set Application Date *API Name: Set_Application_Date

Description:

Set Variable Values

Each variable is modified by the operator and value combination.

Variable	Operator	Value
...application_c > Application Date	Equals	...ng Flow Interview > CurrentDate

+ Add Assignment

Test 2

Created another Application record.

Result: An email notification was sent automatically.

Search...

Sales Home Opportunities Leads Tasks Files Accounts Contacts Campaigns APP-0001 | Application New Contact Edit New Opportunity

Application APP-0001

Related Details

Application Name	APP-0001	Owner	Kaligotia Sri Datta Sai Vithal
Student	Datta		
Job	Frontend web developer		
Application Date	30/07/2026		
Status	Applied		
Created By	Kaligotia Sri Datta Sai Vithal, 30/07/2026, 11:52 am	Last Modified By	Kaligotia Sri Datta Sai Vithal, 30/07/2026, 11:57 am

Activity

Filters: All time • All activities • All types Refresh Expand All View All

Upcoming & Overdue

No activities to show.

Get started by sending an email, scheduling a task, and more.

To change what's shown, try changing your filters.

Show All Activities

To Do List

Test 3

Verified that both Flows were active and executed successfully.

The screenshot displays the Salesforce Flow Builder interface. The top navigation bar includes the 'Developer Edition' label and a 'Flow Builder' tab. The main workspace shows a flow diagram with the following steps:

- Record-Triggered Flow** (Start): Object: **Application**, Trigger: **A record is created**, Optimize for: **Actions and Related Records**.
- Run Immediately** (Intermediate step).
- Send Email** (Action): Configured with the following details:
 - Subject**: New Application Submitted
 - Body**:

```
Hello Placement Officer,  
  
A new application has been submitted.  
  
Student: {!$Record.Student__r.Name}  
Job: {!$Record.Job__r.Name}  
Status: {!$Record.Status__c}  
  
Please login to Salesforce to review the application.
```
- End** (Final step).

The right-hand pane shows the configuration for the 'Send Email' action, including options to 'Compose Email Content' or 'Use Email Template', and a rich text editor for the email body.

Challenges Faced

- Initially tried using the Run button to test a Record-Triggered Flow.
- Learned that Record-Triggered Flows only execute when a record is created, updated, or deleted.
- Understood the difference between Before Save and After Save Flows.
- Learned how to use related record fields inside the Send Email action.

Key Learnings

- Understood when to use Validation Rules, Flows, and Triggers.
- Learned to build Record-Triggered Flows.
- Learned to use Assignment and Send Email actions.
- Understood the difference between Before Save and After Save Flows.
- Gained confidence in choosing the correct automation tool for different business requirements.

Conclusion

Day 3 helped me understand Salesforce automation in a practical way. I learned how to automate business processes using Record-Triggered Flows, validate data using Validation Rules, and identify situations where Apex Triggers are required. I also implemented automation in the Placement Management System and tested the flows successfully.